1 Purpose

This document describes the approach to computerized system validation within SCHOTT Pharma and is applicable for internal suppliers and is a guidance for external suppliers.

Rather than to guarantee a fixed quality level, this SOP enables SCHOTT Pharma and the suppliers to

- have a common understanding of the sourcing procedure.
- perform a risk-based approach according to GAMP 5 to bring new software into an office or production environment at SCHOTT Pharma sites.
- assess the quality of software and computerized systems.
- perform a fact-based decision process.

This SOP is in accordance with the P Validation and Qualification Guideline [01] and covers the requirements of the Data Integrity Guideline for Computerized Systems [02].

SCHOTT Pharma shall define the relevant computerized systems [12], which shall follow the rules given and shall be validated according to this SOP. The definition and classification of the OT system is given by the SOP Management of OT Systems.

2 Scope

This IT Validation Guideline is in effect for all validation activities of software and computerized systems in all departments of SCHOTT Pharma. New validation activities for computerized systems shall comply with this guideline.

Not in scope:

This SOP is not intended for the assessment of “Software as Medical Device” products.

Document release

There can be more than one approver. They are not displayed here, but recorded in the DMS.

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Table of Content

1 Purpose 1
2 Scope 1
3 Introduction 4
4 Responsibilities 5
4.1 Supplier related responsibilities 6
5 Abbreviations and Definitions 7
6 Procedure 8
7 General Requirements 10
8 Qualification & Validation Strategy 10
9 Qualification Planning 10
9.1 Risk Assessments 11
9.2 Hardware Requirements 11
9.3 Software Requirements 11
9.4 Software Integration Requirements 12
9.5 Software Release 13
9.6 Qualification utilizing GAMP 5 Categories 13
9.7 Recommended Activities Based on GAMP Categories 14
10 User Requirement Specification (URS) 16
10.1 Supplier’s response to the URS: Functional Specification or similar specification documents 16
11 Testing 16
12 Design qualification (DQ) 17
13 Installational Qualification, Operational Qualification, Performance Qualification (IQ, OQ, PQ) 17
13.1 Installational Qualification (IQ) 17
13.2 Operational Qualification (OQ) 17
13.3 Performance Qualification (PQ) 18
14 Qualification Report 18
15 Validation planning and (Process) Validation 18
16 Validation Report 18
17 Traceability Matrix 19
18 Execution and Monitoring 19
19 Operation and maintenance of qualified status 19
20 Input/ Output/ KPIs 19
21 Training 19
22 Applicable standards 20

23 Reference Documents.....20

24 Document History.....21

3 Introduction

A computerized system per ISPE GAMP 5 Guide is defined as software and hardware components, together with their controlled process or function (see Figure 1).

Each part of a computerized system needs to be appropriately tested before entering a manufacturing environment of SCHOTT Pharma. Furthermore, software components need to be integrated into the hardware components and the software environment. Additionally, to testing each component, the integration of software components also needs to be tested.

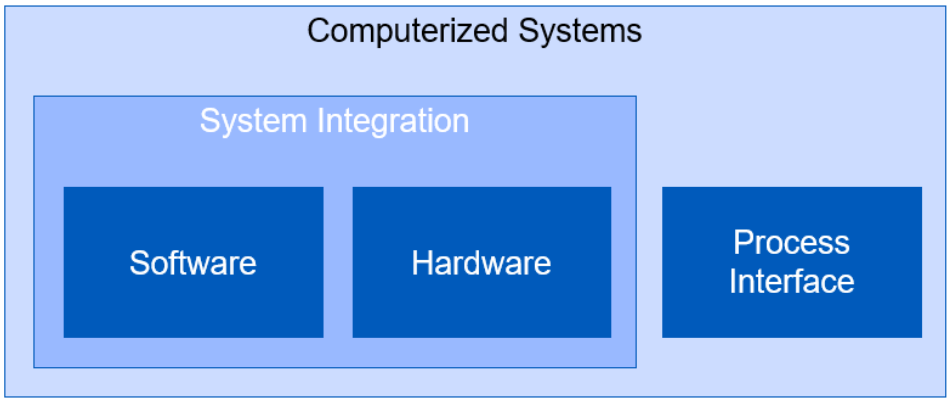

Figure 1: Computerized systems overview

The release and deployment of the software bears its own challenges and often includes many manual steps. To reduce the potential for errors, it shall be possible to install the software in a safe and repeatable manner - for example by provision of an installer or image file (Software Deployment Package has been established as broader term). The Software Deployment Package and installation instructions have to be tested as well.

The validation engineer usually doesn't have the expertise to plan or perform most of those activities. The validation engineer will collect these testing deliverables from the supplier. The expertise lies only within the development team of the software. In many cases, the development team is not available to produce any deliverables for the project. Especially in case of a commercial off the shelf software, it is hard to get any information on the product. For this reason, it is important to assess the documentation gaps and estimate the risks connected to those.

In general, it is less expensive to detect errors at an earlier stage (see Figure 2). All these measures are taken to detect errors as early as possible and avoid a cost overrun.

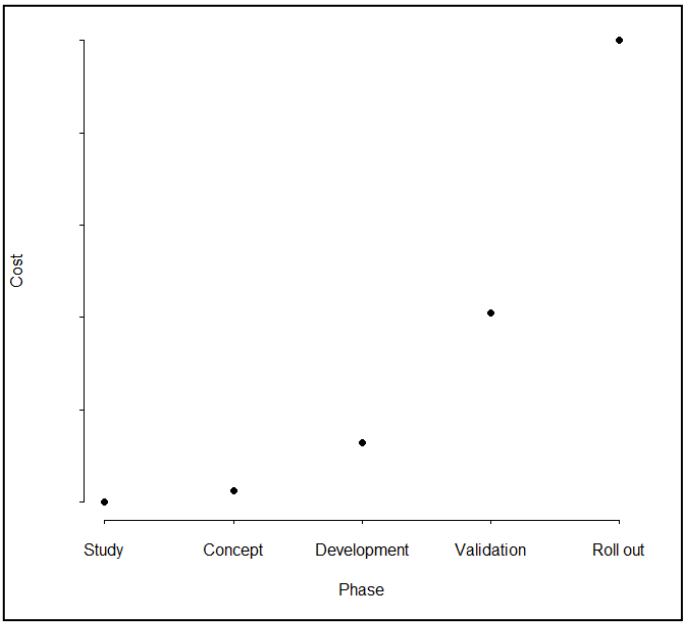

Figure 2: Error cost escalation

SCHOTT Pharma will perform qualification and validation activities on production equipment - including computerized systems - according to the needs identified in a risk analysis. All documents ordered from suppliers for this purpose will be referenced in a SCHOTT Pharma framework document (e.g. VMP). With this approach, an integrated engineering is ensured.

For computerized systems all software related topics are covered within this SOP. Physical components of computerized systems and the complete production line and their environment are covered in SOP Equipment Qualification and Process Validation [04].

To establish a baseline for the expected quality regarding computerized systems, SCHOTT Pharma has defined the following sections.

4 Responsibilities

The responsibilities related to this SOP are described in the table below.

Table 1: Responsibilities of this SOP

Responsibility	Function/ department
Owner of procedure	Head of Quality Systems and Processes
Procedure team	Validation Engineer Responsible User Execution Staff local / global Quality representatives System Owner Process Owner

The necessity for a computerized system validation can be identified by various sources – for example from a capacity expansion project, a change request or a ServiceNow ticket.

The responsibilities shall be defined for each project and integrated into the Qualification and Validation documentation. The following table (Table 2) gives an example of typical roles and the functions responsible for those.

Table 2: Responsibilities (**Example**)

Responsibility	Function/ department
Responsible User	Head of Production
Execution Staff	Production staff of the line
Validation Coordinator	Validation Engineer
Quality representative	Site Quality Manager
System Owner	Global Quality Manager
Process Owner	Process Engineer

The roles named in the Table 1 are defined in the SOP Validation and Qualification Guideline [01] and SOP Data Integrity Guideline [02].

4.1 Supplier related responsibilities

For systems changed or created at a supplier by request of SCHOTT Pharma and to identify differences in the development procedures for this purpose, the following subchapters establish the expected cooperation interfaces of a supplier and SCHOTT Pharma.

4.1.1 Software developed by suppliers

SCHOTT Pharma expects that each supplier developing software works with a gated development process to make sure the delivered software or computerized system attains to the agreed quality level. At the end of the software development process, SCHOTT Pharma expects the supplier to follow a formal software release process. All changes have to be tracked in a Change Log or similar structured format, containing at least the change number and a description of the implemented changes. Maintaining such a structured change list does not release the supplier from his obligations to update the software documentation.

SCHOTT Pharma assumes that the supplier's software release process consists at least of:

- Static code analysis
- Unit Testing
- Integration Testing
- Acceptance Testing

A summary of each of these phases and a list of documents have to be provided to SCHOTT Pharma upon request and are accepted as Software Release (see 9.5).

4.1.2 Hardware provided by suppliers

SCHOTT expects that each supplier providing hardware fulfills the requirements listed in 9.2 and is ready to hand over the corresponding documentation at the agreed project milestone.

4.1.3 Software sourced by suppliers (COTS and SOUP)

SCHOTT Pharma expects that each supplier sourcing software or using pre-existing code libraries (such as .NET Frameworks), creates a list of the used versions. This is necessary for identifying IT security concerns. Additionally, the supplier fulfills the requirements listed in 9.3 and is ready to hand over the corresponding documentation at the agreed project milestone or to at least show it per request during a supplier audit.

4.1.4 Other supplier deliverables

Additionally, to providing the software product or system SCHOTT Pharma has ordered, the supplier has to formulate a solution approach (Functional Specification, see also10.1) for each requirement listed in the URS. The solutions shall be based on a risk analysis, planned together with SCHOTT Pharma and reviewed by SCHOTT Pharma. Other deliverables may be agreed upon per project.

5 Abbreviations and Definitions

For a list of abbreviations and definitions relevant for SCHOTT Pharma, please see 'List of Abbreviations for SCHOTT P Unit' (Doc ID: 10000410346).

Abbreviations specifically relevant for this document are listed in the table below.

Table 3: Abbreviations & Definitions

Abbreviation / Term	Explanation
Acceptance Test	Formal test with pre-defined acceptance criteria to form a conclusion whether the software is accepted or not.
Code Review / Static Code Analysis	Independent review of the written code by another developer or automated system.
Configurable Software	(Commercial) off-the-shelf software that can be configured for specific user applications without altering the basic program
Configuration	state-specific modifications related to the part of the software which is not the source code (software binaries) or changes of structural data model
COTS	Commercial off the shelf software.
CSV	Computerized System Validation (See section 3 for further definition).
Integration Test	Formal test of the software within the intended environment.
GAMP	Good Automated Manufacturing Practice.
Regression Test	Test to ensure that the previously existing functionality was not affected.
Software Deployment Package	Package of the software and all required components, third party software and libraries.
SOUP	Software of unknown provenance, for example code libraries or open-source software.
Unit Test	Test function to test a specific part of the source code.

6 Procedure

Table 4: Process flow

Supplier	Input	Process	Output	Customer
SCHOTT Pharma	Requirements Software and Hardware	Describing requirements	URS released	Internal or external supplier
Internal or external supplier	Released URS	Performing risk assessment	Risk Assessment released	SCHOTT Pharma (On request)
Internal or external supplier	Released URS Released Risk Assessment	Assessment of requirements	Functional Specification released	SCHOTT Pharma (On request)
Internal or external supplier	Functional Specification released	Software development	Functional complete software	SCHOTT Pharma (On request)
Internal or external supplier	Functional complete software	Software Unit testing	Unit test report	SCHOTT Pharma (On request)
Internal or external supplier	Functional complete software	Software Integration testing	Integration test report	SCHOTT Pharma (On request)
Internal or external supplier	Functional complete software	Software Acceptance testing	Acceptance test report	SCHOTT Pharma (On request)
Internal or external supplier	Successful testing	Creation of final software version	Software Release	SCHOTT Pharma (On request)
SCHOTT Pharma	Software Release	Performing DQ	DQ report	SCHOTT Pharma
SCHOTT Pharma	Successful released DQ	Performing IQ at the fully equipped production line, (if applicable) testing interfaces at the production line, (if applicable)	IQ reports released	SCHOTT Pharma
SCHOTT Pharma	Successful released IQ	Performing OQ	OQ reports released	SCHOTT Pharma
SCHOTT Pharma	Successful released OQ	Performance testing	Performance test report	SCHOTT Pharma

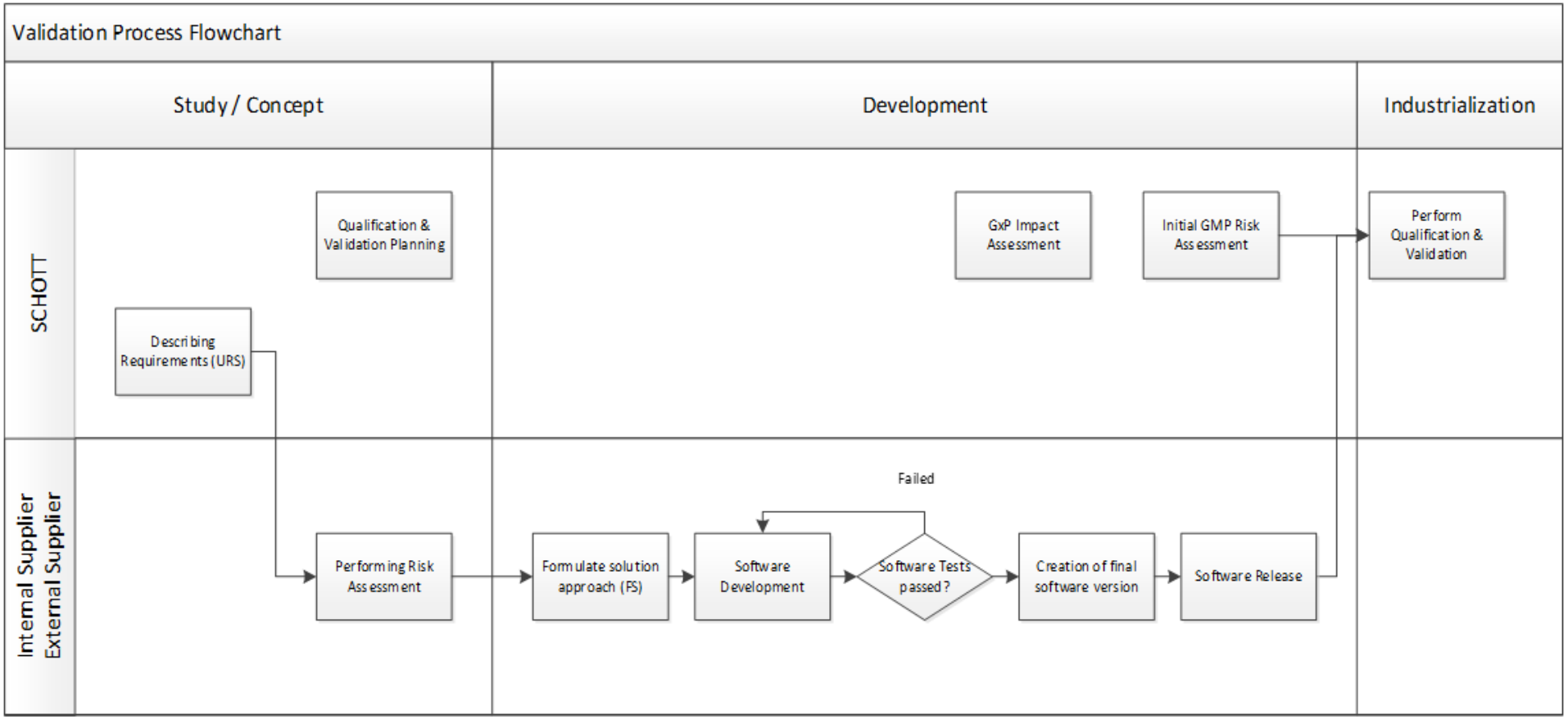

Figure 3: Process flow

7 General Requirements

Computerized systems that fall within the scope of this SOP need to be validated before being used for production work. This procedure is applicable for all new computerized systems and changes to existing computerized systems. For OT related systems, this definition is given by the process defined on the SOP P IT Management System [06].

Validated computerized systems shall have the following characteristics:

- developed by qualified personnel according to a pre-defined development procedure which produces pre-defined deliverables
- protected from unauthorized modification
- satisfy user requirements
- supported and maintained by qualified personnel in accordance with change control procedures
- utilized by trained and qualified users
- involved management approval of the validation process
- validation deliverables reviewed and approved by qualified personnel

8 Qualification & Validation Strategy

The user needs for a Computerized Systems shall be collected and comprehensively described in a User Requirements Specification [03]. Each requirement has to be described in a way, that specific quality criteria can easily be derived. Each requirement shall be identified and tracked until the validation is closed. The implementation of all relevant requirements shall be followed up upon.

The supplier shall specify the properties and architecture of the Computerized System in one or more technical documents (e.g.: Functional Specification – higher level document explaining the architecture, or Technical Specification – tables, libraries, GUIs etc.), based on those requirements. SCHOTT Pharma may review, request changes (if necessary) and may sign the proposed technical documents. SCHOTT Pharma shall classify the software according to the GAMP 5 approach and Data Integrity Guideline for Computerized Systems [02].

The classification of the software is done with the GxP Impact assessment [03]. This classification considers the criticality of the data processes and the system categories as defined in GAMP 5. From a risk perspective it is expected that the qualification and validation efforts are proportional to the classification of the system. The strategy depends on the kind of system being implemented or changed.

After the Computerized System is built, tested and released by the supplier, SCHOTT Pharma collects technical documents and testing summaries from the supplier and complements these with additional risk-based tests, if necessary. The test protocols can be combined with the SOP Equipment Qualification and Process Validation [04] activities, starting with a planning document.

9 Qualification Planning

The qualification needs to be executed in a systematic approach. For this reason, a Qualification Plan (Validation Plan) is created [03]. The Qualification Plan defines the scope and the strategy for the qualification project, the hierarchy of documentation and the responsibilities.

Risk management is always part of qualification and validation – it is either planned or referenced in the Qualification Plan – and is the foundation for creation of the qualification test protocols. In general, each

GMP requirement shall be tested at the earliest possible moment. For this reason, the bulk of tests should be executed in the Design Qualification phase.

The following sections give minimum requirements for each qualification. If gaps are identified, they shall be assessed within the project.

9.1 Risk Assessments

The following types of risk assessments are applicable:

- GxP Impact Assessment: Provides classification of the software according to GAMP 5 and SOP Data Integrity for computerized Systems [02]
- GMP Risk Assessment: Provides test coverage of all User Requirements. See SOP Equipment Qualification and Process Validation [04].
- FMEA: Provides risk mitigation during the entire product life cycle. See SOP FMEA Procedure [05].

9.2 Hardware Requirements

As a main prerequisite for validation of computerized systems, the hardware hosting the application software shall be qualified. At least the following requirements for the hardware and all its components shall be fulfilled.

The hardware / hardware components…

- Are clearly identified in documents and on the physical units. If necessary, identification plates must be attached to the components.
- Are part of a Bill of Materials and are inspected on arrival.
- Have written specifications, including interfaces and network connections, and are fit for their purpose.
- If necessary, need to be calibrated and labelled with a calibration sticker.
- Fulfil the requirements of the installation site (e.g., clean room).
- Are installed according to the specifications.
- Are qualified before software testing and integration testing. Qualification may be repeated at the final installation site.

Those are additional requirements for computerized systems, to the equipment requirements derived from SOP Equipment Qualification and Process Validation [04].

9.3 Software Requirements

All application software, the software / software components…

- Are clearly identified with name and version.
- Have written specifications, including interfaces and network connections, and are fit for their purpose.
- Are sourced directly from the originators deployment system or the official distributors.
- Are installed according to the specifications.
- Deployment within SCHOTT are tracked in a tracking system or list.
- Are checked or tested with the specifications or installation instructions.
- Have build notes, release notes, a change log, or a similar format to inform about changes.

- Have a list of the included sourced software (SOUP – Software Of Unknown Provenance), if applicable.

Those and further requirements, such as functionality, translations and specification of the user interface have to be defined in the User Requirement Specification, as appropriate.

9.4 Software Integration Requirements

All application software must be properly integrated in their hardware and software environment. For this purpose…

- The firmware version for each component needs to be clearly identified with name, version and origin.
- The functionality/responsiveness of each hardware component is shown by a simple test or by systematic test runs.
- It is demonstrated that the system can safely shutdown and restart according to the requirements of the project. Especially concerning data persistence, log files and Audit Trail in emergency procedures or alarm states.
- Maintenance and service procedures are working as intended.
- Expected (handled) error states are tested and described in a table.

Those and further requirements, if necessary, have to be defined in the User Requirement Specification, as appropriate.

9.5 Software Release

After the software development is finished, a release statement for the software shall be given. The Software Release includes:

- Final software version number.
- Type of the Software Deployment Package.
- Distribution method.
- Summary of the performed testing.
- Latest Change Log file or similar.

If this has to be done formal or informal and by SCHOTT Pharma or the supplier is decided per project. The Software Release contains the base information for the Design Qualification Report.

Note: The start of the distribution is a major milestone (earliest reached after the DQ) and has to be planned in the project and approved by SCHOTT Pharma.

9.6 Qualification utilizing GAMP 5 Categories

SCHOTT Pharma IT systems are classified in accordance with GAMP 5. This classification shall be used to decide the extent of the qualification and validation efforts for the system. The sections below provide detailed Guidance on that.

Within the Data Integrity Guideline [02] it is shown how the GAMP 5 software categories can be used to explain the qualification and validation extent. Those categories classify software into groups according to their associated risk. The classification of the software is done with the GxP Impact assessment [03].

Please note that the following exemplary qualification activities derived from the GAMP 5 categorization do not substitute a thorough analysis of the associated risk. The qualification activities are always determined by the risk analysis team members and are based on the associated risk of the failure mode or requirement in combination with the GxP classification. The following categories **build upon each other**. For example, category 3 includes all requirements from category 1 and so forth.

9.6.1 Category 1 – Infrastructure software including operating systems, Database Managers, etc.

The required software and/or operating system and version need to be documented within the specifications or installation instructions. For computerized Systems used for production, the presence of category 1 software needs to be checked in the qualification. It is recommended to test the basic functionality of the category 1 software, if possible.

9.6.2 Category 3 – Non-configurable software including commercial off the shelf software (COTS), Laboratory Instruments / Software

The correct installation is verified by functional testing. Interfaces to the hardware and software environment (Integration, e.g., to the active directory) are assessed and tested, if possible. User Management must be documented. Users must be trained on the system. The need for Audit Trail is determined by risk assessment.

9.6.3 Category 4 – Configured Software including LIMS, SCADA, ERP, DMS, etc.

The software configuration needs to be described in a specification document. The software is then tested with those specifications for correct setup, error handling and performance. Audit Trail is required.

9.6.4 Category 5 – Bespoke (own developed) Software

In order to ensure a high probability of success in the qualification and validation activities, it is recommended to develop software on a gated process with sufficient testing and a formal software release procedure. As soon as the Software Release is done between the supplier and SCHOTT Pharma, Design Qualification will be started at SCHOTT Pharma immediately.

For bespoke software, it is required to provide documents explaining the software functionalities and operation (e.g., manuals, technical documents). It is also required to provide evidence that the development documentation is available to allow proper software maintenance, such as source code and configuration descriptions. Such documents shall be maintained in a controlled way.

For category 5 it is required in the qualification and validation project to present documents explaining the software functionalities and operations. Besides that, it is recommended to verify if the software was developed according to the supplier's software lifecycle development procedure. This is done through auditing.

9.7 Recommended Activities Based on GAMP Categories

Table 5 provides a list of recommended activities for the qualification and validation based on GAMP categories. The Validation Coordinator and team shall decide which qualification and validation activities are going to be executed as part of the project. This shall be documented on the Qualification/Validation Plan or subsequent documents. This decision should also take into consideration the data category [02] associated to the systems use cases.

Table 5: Recommended activities based on GAMP categories

	Hardware	Category 1	Category 3	Category 4	Category 5
System Assessment					
Location of the system and its environment	X	-	-	-	-
Installation of hardware and verification of components	X	-	-	-	-
Software installation and verification of components	-	X	X	X	X
System documentation:					
System configuration document	X	X	X	X	X
Manuals	X	X	X	X	X
Specifications	X	-	X	X	X
System-Tests:					
Start up and shut down	(X)	X	X	X	X
System availability and emergency concept	(X)	(X)	(X)	X	X
Access and security	(X)	(X)	(X)	X	X
Application and function tests:					
Data Input and boundary conditions	-	-	-	X	X
Display and reports	-	-	X	X	X
Messages and alarms	-	-	-	(X)	(X)
Decision points	-	-	-	X	X
Critical algorithms and calculations	-	-	(X)	X	X
Data storage and access	-	-	(X)	X	X
Data backup and restore	-	-	(X)	X	X
Data changes and audit trail	-	-	(X)	X	X
Stress conditions	-	-	(X)	X	X
System interfaces and data exchange	-	(X)	X	X	X
Data archiving and retrieval	-	-	(X)	X	X
User requirements tests	-	-	(X)	X	X
Operating environment SOPs:					
Use and Operations	X	(X)	X	X	X
Backup and restore	-	-	(X)	X	X
Archival and retrieval	-	-	(X)	X	X
Access and Security	X	(X)	(X)	X	X
Disaster recovery plan, emergency concept	X	(X)	(X)	X	X
Decommissioning	-	-	(X)	X	X
System supporting SOPs:					
Configuration management	X	-	-	X	X
Maintenance	X	X	X	X	X

X = recommended / (X) = recommended, if applicable

10 User Requirement Specification (URS)

The requirement specification describes the user requirements by a Top-Down-approach and implies quality criteria at an early stage. Each requirement should be identifiable clearly by a number, complete, realistic, without antagonism and testable.

Each requirement should be prioritized into two categories:

- GEP: Requirements only important for developing the system.
- GMP: Requirements with impact on produced product quality, data integrity.

In general, the requirements from section 9 shall be covered.

10.1 Supplier's response to the URS: Functional Specification or similar specification documents

The supplier outlines a detailed description of the implementation planned by the developer in a way that both the system owner and the developer can understand and in a way that nothing is left unclear. Each requirement of the URS shall be covered and identified by a unique ID number. Those specifications are grouped into viable subsystems (for example modules) and their interaction is plotted on a high level. Each subsystem/module must be assigned a clear name based on which it can be identified. Diagrams and graphs should also be used if possible and helpful to illustrate information.

11 Testing

Testing is in the responsibility of the supplier as far as possible. To define the extent of testing, risk assessment shall be done. In general, the produced code shall be reviewed by a qualified person or automated software (Static code analysis) and automated tested to prevent unforeseen changes (Module/Unit Tests), if possible. The integration needs to be tested (Integration Testing), the functional aspects need to be tested with Acceptance Testing and possible impacts on pre-existing code needs to be covered with Regression Testing.

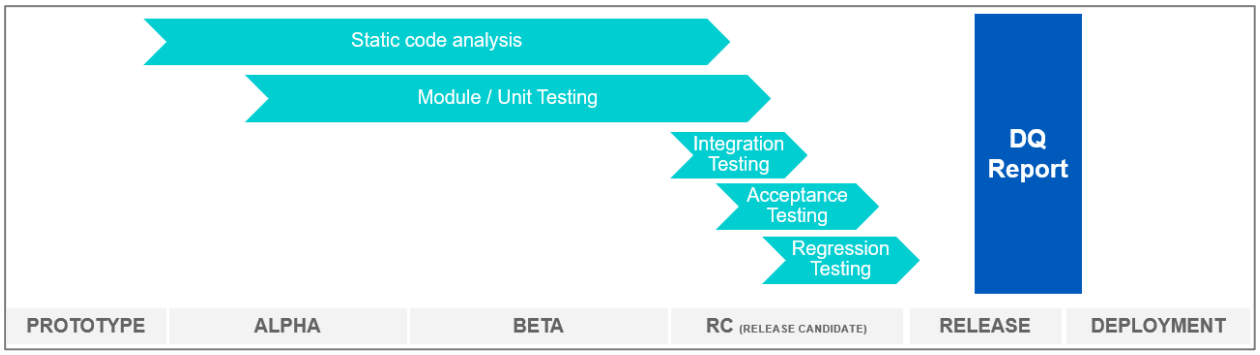

Figure 4: Software development phases, testing and qualification

The supplier might have additional requirements, to the requirements from SCHOTT Pharma, for the hardware. Hardware components shall be inspected for their suitability.

Detailed summaries of the tests executed shall be released according to Good Documentation Practice and collected for the DQ, together with the required proofs for the requirements in section 9.

12 Design qualification (DQ)

Testing reports (or summaries) of the Release Candidates and the Software Release documentation is collected and assessed within the Design Qualification. Deviations and modifications of the system specification during development and testing are documented and commented within the DQ. In cases, where these documents cannot be retrieved for the project, the risk shall be assessed and if the risk is deemed acceptable, the Software Release documentation (section 9.5) can be provided by SCHOTT Pharma.

SCHOTT Pharma is always part of the DQ release process. With the successful Design Qualification Report, the Software Deployment Package is released for deployment. Means, that it is ready for the FAT/SAT and finally the delivery to SCHOTT Pharma or be installed on SCHOTT Pharma production equipment via Change Request.

13 Installational Qualification, Operational Qualification, Performance Qualification (IQ, OQ, PQ)

After the successful Design Qualification (DQ) and – depending on the project – FAT/SAT, the computerized system can be brought into final location in the production environment. While the DQ is typically done once and is only repeated after changes, the Installation Qualification and Operational Qualification is done for each installation. The corresponding IQ, OQ and PQ tests for computerized systems are derived from the risk analysis and are incorporated into the equipment qualification with SOP Equipment Qualification and Process Validation [04].

13.1 Installational Qualification (IQ)

When the system is installed in the final software and hardware environment, an Installation Qualification shall be carried out. The aim of the Installation Qualification is to demonstrate, that <u>each</u> installation and configuration has been carried out correctly and no issues with the immediate environment exist.

For example:

- Software / Hardware installation checks. Identification.
- Shutdown and restart of the system with no data loss.
- User management interface with the active directory of the site is working.
- Network/ Server access is working as required.
- Safety aspects, (warning) labels and user manual.
- Other tests as defined in the risk analysis.

The successful completion of the IQ is the precondition to execute the Operational Qualification (OQ).

13.2 Operational Qualification (OQ)

After the successful IQ, the functional and process aspects of the installed system are tested. The aim of the Operational Qualification is to demonstrate, that the system is working within the specified limits and the work environment is ready for operation.

For example:

- Configuration of batches/ test setups. Successful execution of setup wizards.
- Nominal production is possible with no critical errors.
- Test runs at the process limits.

- Other tests as defined in the risk analysis.

The successful completion of the OQ is the precondition to execute the Performance Qualification (PQ).

13.3 Performance Qualification (PQ)

Within the PQ, the system is evaluated in production over a duration, with nominal settings. Therefore, the expected outcome are good products. It is possible, that the products are delivered to the customer. The required quality checks for a delivery shall be documented and the release procedure of those products has to be defined. The production parameters of the batches shall be documented.

For example:

- SOP is released and trained.
- Setup for the batch was done correctly.
- Production of the required samples / over the required duration (e.g., 3 batches over 1,5 shifts each).
- Inspection of the products.
- Other tests as defined in the risk analysis.

14 Qualification Report

The Qualification Report summarizes and closes the qualification phase with a qualification decision.

15 Validation planning and (Process) Validation

A validation plan shall always be established before the validation phase. The validation plan usually is established within SOP Equipment Qualification and Process Validation [04].

The validation plan includes a description of the process that shall be validated, documents concerning the procedure in the project, planned validation documents, as well as relevant SOPs, changes in the course of the project and further documents. If validation requirements derive from the software, the corresponding documentation regarding those requirements and the test strategy need to be incorporated into the validation plan and test protocol.

The validation test protocol defines which validation steps are carried out in which extent and their acceptance criteria. Whenever possible, documented evidence (screenshots, files) shall be created to prove the acceptance criteria for software was fulfilled.

For example:

- The system was developed under the right premises.
- The produced output is within the specification of the product over the complete parameter range.
- Evaluation / Confirmation of inspection performance.

16 Validation Report

The Validation Report summarizes and closes the validation phase with a validation decision.

17 Traceability Matrix

The Traceability Matrix is a link between risk analysis, specifications and test protocols. The Traceability Matrix enables a systematic testing and guarantees that the GMP User Requirements are addressed completely.

18 Execution and Monitoring

The qualification and validation shall be done in parallel to the project. All documents must be signed, approved, and released. The qualification and validation are concluded by the individual in charge of the validation. The rules of change control must be observed.

19 Operation and maintenance of qualified status

All components are subject to a formal change management starting with the go-live. Changes are documented, evaluated and approved before they can be released.

20 Input/ Output/ KPIs

No specific KPI is defined for the validation and qualification process.

21 Training

Qualification and validation shall be carried out by qualified personnel. All personnel involved in the creation, approval and/ or release of qualification and validation documents [03] and/ or in qualification and validation documentation has to be appropriately trained.

In order to do so, the following training requirement lists are provided in CompliantPro™:

- Validation execution staff training requirement list [07]
- Validation and Qualification Coordinator [08]
- Validation and Qualification Expert [09]
- Validation and Qualification Responsible user [10]
- Validation and Qualification Quality representative [11]

Project managers shall ensure that the internal project participants are trained in accordance with the training requirement lists above, in connection to their roles in the project.

In addition to that, the performing personnel shall be trained to the test protocols, in particular to the test instructions, if activities are performed that go beyond the scope of the employees' usual routine procedures. This is in responsibility of the creator of the test plans.

22 Applicable standards

Table 6: Applicable standards

No.	Document ID	Explanation
01	10000015495	ISO 15378 Primary packaging materials for medicinal products — Particular requirements for the application of ISO 9001:2015, with reference to good manufacturing practice (GMP)
02	10000403048	ISO 31000: Risk management — Guidelines
03	n/a	EU GMP Annex 11: Computerized Systems
04	n/a	GAMP® 5 „A risk-based approach for GxP-compliant computerized systems “
05	n/a	FDA Regulation 21 CFR Part 210, 211 (US-GMP)
06	n/a	FDA Regulation 21 CFR Part 11 “Electronic Records; Electronic Signatures”
07	n/a	EU Guideline for GMP

23 Reference Documents

Table 7: Reference documents

No.	Document ID	Explanation
01	10000007736	P Validation and Qualification Guideline
02	10000393773	Data Integrity Guidelines for Computerized Systems
03	10000040057	Template Overview
04	10000453255	SOP Equipment Qualification and Process Validation
05	10000013961	SOP FMEA Procedure
06	10000056220	SOP Management of OT related Systems
07	TRL-00001 / 4	Validation execution staff training requirement list
08	TRL-00024 / 0	Validation and Qualification Coordinator
09	TRL-00025 / 0	Validation and Qualification Expert
10	TRL-00026 / 0	Validation and Qualification Responsible user
11	TRL-00027 / 0	Validation and Qualification Quality representative
12	10000046099	Inventory List of standard OT Systems

24 Document History

Table 8: Document history

Version	Date of change	Change in chapter xy / description of change / justification of change
0.1	2008-07-11	1. version of the valid document
0.2	2009-05-31	2. version of the valid document
01	2010-12-28	New Document No. and Versioning in SAP DMS/ Handling of old existing (not validated) Systems
02	2013-01-03	Separation and Reference to Template Document
03	2014-06-30	Reference to updated Templates & minor error corrections
04	2018-01-31	Revision with respect to data integrity, version not released
05	2018-03-02	Revision by PP-T (Meinhardt, Rauscher, Scheller)
06	2022-06-10	Update of SOP due to regular update interval: • Transfer into new template version 10000015137/ 11. • Review and adjustment of templates in reference [03]. • Excluded “Software as medical device” in scope. • Updated Responsibilities according to changed templates. • Added new Definitions and removed duplicates with 10000410346. • Included explanation to section 15. • Merged Sections Archiving and Documentation. • Adjusted process according to changed references.
07	2023-03-20	Complete overhaul of the SOP for segregation of testing and qualification activities. Small adaption due to organizational re-structuring of PP-Technology within global function “R&D and Global Engineering Pharma” with March 1 2023.